Experimental Gerontology 214 (2026) 113022

Contents lists available at ScienceDirect

Experimental Gerontology

journal homepage: www.elsevier.com/locate/expgero

The effects of paroxetine on lifespan, feeding behavior, and other physiological parameters in the nematode *C. elegans* under a modified bacterial diet

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ARTICLE INFO

Section Editor: Ricki Colman

Keywords:
Paroxetine
Caenorhabditis elegans
Lifespan
Feeding behavior
Fat metabolism
Bacterial consumption
High-glucose diet

ABSTRACT

Paroxetine, a widely prescribed selective serotonin reuptake inhibitor (SSRI), is known for its metabolic side effects, yet its long-term physiological impacts remain incompletely understood. This study investigated the effects of paroxetine on lifespan, feeding behavior, and fat accumulation in *Caenorhabditis elegans* across different developmental stages and dietary conditions. We found that low-dose paroxetine extended lifespan when exposure began early, while high doses consistently reduced lifespan. Notably, the lifespan-extending effects appeared to be diminished when worms were fed a high-glucose diet, suggesting that dietary context may influence the physiological outcomes of paroxetine. Paroxetine also increased food intake without causing fat accumulation, indicating a possible metabolic uncoupling. These effects were independent of major serotonergic, dopaminergic, and insulin-like signaling pathways, pointing to alternative mechanisms. Overall, our findings highlight the potential roles of dose, timing of exposure, and diet in shaping the physiological impacts of SSRIs.

1. Introduction

Prescriptions for Selective Serotonin Reuptake Inhibitors (SSRIs) abound to treat anxiety and depression (Fluyau et al., 2022). Though most people consider SSRIs to be safe and efficient, long-term use of them has been linked to a variety of undesirable side effects, including weight gain, metabolic difficulties, cardiovascular diseases, and changed glucose metabolism (Bansal et al., 2022; Danborg et al., 2019; Cartwright et al., 2016). Among SSRIs, paroxetine, in particular, has been linked to higher hunger, fat storage, and lower insulin sensitivity, which raises questions regarding its possible contribution to obesity (Nevels et al., 2016). Beyond metabolic effects, SSRIs like paroxetine also impact neurodevelopment, endocrine regulation, and mitochondrial function, indicating broader systemic influences beyond neurotransmitter modulation (Gill et al., 2020; Ranjbar et al., 2013; Nihalani et al., 2011; Edinoff et al., 2021).

These adverse effects have been associated with an elevated risk of mortality in both animal models and human studies. Long-term SSRI use has been correlated with increased all-cause mortality, particularly in older adults, potentially due to metabolic dysfunction, cardiovascular stress, and drug interactions (Ön et al., 2022; Prasitlumkum et al., 2021; Zivin et al., 2016). In animal models, chronic SSRI administration has also been linked to metabolic dysfunction and cardiovascular stress, which leads to reduced survival rates, indicating that serotonergic modulation may influence longevity through mechanisms beyond mood regulation (Kummet et al., 2012; Haskell et al., 2013; Lu et al., 2024; Hammer et al., 2025; Gaukler et al., 2015; Karanges et al., 2011; EL-Gaafarawi et al., 2005). The relationship between SSRIs use and longevity is complex and not yet fully understood. While some studies point to certain effects on lifespan and health span, the biological mechanisms behind these effects remain unclear. Ongoing research is needed to explore how SSRIs interact with key physiological pathways involved in aging.

The model organism *Caenorhabditis elegans (C. elegans)* has proven to

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https://doi.org/10.1016/j.exger.2025.113022
Received 23 August 2025; Received in revised form 29 December 2025; Accepted 31 December 2025
Available online 5 January 2026
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be an excellent system for studying aging due to its short lifespan, well-characterized biology, and fully sequenced genome (Tissenbaum and Using, 2015; Jeayeng et al., 2024). The potential impact of paroxetine on lifespan in *C. elegans* has been the focus of multiple investigations, yielding varied observations. An early study reported that high doses (50–250 μM) of paroxetine shortened lifespan when exposure began in adulthood, suggesting that the drug may negatively impact longevity at high concentrations (Petrascheck et al., 2007). Moreover, another study has demonstrated that exposure to high doses of paroxetine during the early embryogenic phases proved toxic to *C. elegans*, suggesting that the timing of treatment initiation may greatly influence the physiological outcomes related to the drug (Weeks et al., 2018). In contrast, more recent work has indicated that low doses (0.1–1 μM) of paroxetine can extend lifespan, whereas high doses of the drug shorten lifespan (Zhou et al., 2024). Their findings suggest that the drug affects lifespan in a dose-dependent manner. They also reported that paroxetine extends lifespan and healthspan in *C. elegans* by modulating the *ser-7* and *dop-4*–IIS pathways (Zhou et al., 2024). However, in this recent study, the time of exposure was not explicitly mentioned, making it essential to consider whether the time of exposure would alter these outcomes. These investigations are part of larger efforts aimed at discovering chemical substances that can promote increased lifespan in *C. elegans*, with the ultimate objective of applying these findings to humans.

However, a crucial element that has been overlooked in prior studies regarding the impact of paroxetine on the lifespan of *C. elegans* is diet. While *C. elegans* mainly feed on a high-protein bacterial diet (Smith, 1992), humans tend to have a high-carbohydrate diet (Shan et al., 2019; Lee et al., 2018; Clemente-Suárez et al., 2022), which could significantly affect metabolic reactions to pharmacological treatments. Understanding how dietary composition influences the longevity effects of paroxetine is crucial for evaluating its potential relevance beyond the nematode model. Additionally, previous studies have not adequately explored other metabolic changes that might contribute to premature death associated with SSRIs, such as changes in appetite and obesity. These factors could play a significant role in the observed outcomes and warrant further investigation to fully understand the complex interplay between paroxetine, diet, and longevity.

With an emphasis on developmental stage and food composition, we further explored how paroxetine impacts longevity, metabolism, and feeding behavior in *C. elegans*. Various concentrations of paroxetine were administered to the worms at different stages of development; experiments were carried out both with and without a high-glucose diet (HGD) to evaluate potential dietary interactions. This research aimed to determine whether the timing of exposure and dietary content influence the lifespan-extending properties of paroxetine. We also investigate the impact of paroxetine on feeding behavior and fat accumulation, thereby providing greater insight into its metabolic effects.

2. Materials and methods

2.1. Strains and media

The study utilized the wild-type N2 strain as the reference strain throughout the experiments. Additionally, we used several mutant strains to explore the potential mechanisms of paroxetine's effects, including *daf-2* (e1370), *daf-16* (mu86), *ser-1* (ok345), *ser-2* (ok2103), *ser-3* (ad1774), *ser-4* (ok512), *ser-5* (tm2647), and *ser-7* (ok1944), *tbh-1* (n3247), *dop-1* (vs105), *dop-2* (CF2805). It should be noted that the *dop-2* (CF2805) mutant is a quadruple dopamine receptor mutant lacking *dop-1, dop-2, dop-3, and dop-4*. For ease of reference in this manuscript, we refer to this strain as (*dop-2*), although it lacks all four dopamine receptors. The wild-type and all the other mutant strains were obtained from the *Caenorhabditis* Genetics Centre (CGC). During the experiment, all worms were maintained on a standard Nematode Growth Medium (NGM) seeded with the *E. coli* strain OP50 as the food source. They were also incubated at 20 °C. In addition, antibiotics (100 μg/mL ampicillin and 50 μg/mL streptomycin) were added to the media to prevent possible bacterial or fungal contamination.

2.2. Drugs preparation

Paroxetine, commercially available as Paroxat 20 mg (paroxetine hydrochloride) tablets, was obtained from a local pharmacy. Tablets were powdered, dissolved in DMSO, centrifuged, and 0.22 μm-filtered before dilution to remove excipients and ensure experimental consistency while preserving translational relevance. Fresh stock solutions of the drug were prepared in 2 % DMSO and diluted as needed to achieve the desired concentrations (0.2 μM, 100 μM). For the High-Glucose Diet (HGD) treatment, glucose was prepared as a 73 mM solution. This concentration was chosen based on previous studies showing it induces physiological changes in *C. elegans* without causing developmental delays (Lee et al., 2009; Schlotterer et al., 2009; Alcántar-Fernández et al., 2018). The drug treatments were introduced by spotting the solutions onto the surface of the NGM plates, ensuring direct exposure to the worms through their bacterial food source. The ratio of OP50 bacterial lawn to treatment solution was maintained at 1:3, allowing for even drug distribution. Control groups received only distilled water with 2 % DMSO, while the HGD-only groups were treated with distilled water containing 2 % DMSO along with the glucose supplementation incorporated into NGM.

2.3. Food intake

To assess the impacts of paroxetine on food consumption, we utilized two complementary methodologies. This dual-pronged approach enabled a comprehensive evaluation of food intake, incorporating both behavioral and physiological measures to investigate the effects of paroxetine on feeding behavior in *C. elegans*:

- **Pharyngeal Pumping Rate (PPRs):** Following established methods with slight modifications (Raizen et al., 2012), pharyngeal pumping rate (PPR) was used as an indirect measure of food intake by assessing the frequency of pharyngeal contractions in adult worms (day3). Worms were exposed continuously to paroxetine starting from the L1, L3, or Young Adult (YA) stage. Before measurement, worms were starved for 60 min, and their basal pharyngeal pumping (off food) was recorded for 30 s at room temperature using a hand counter. The recorded number of contractions was then multiplied to determine the basal pumping rate in pumps per minute. After starvation, worms were transferred back to food, and the (on food) pharyngeal pumping rate was measured again. Food intake was calculated as the difference between the (off food) pumping rate and the (on food) pumping rate. Each experiment consisted of at least 10 worms per replicate, and all treatments were tested in three independent experiments to ensure reproducibility.
- **Bacterial Consumption Rate:** Following established methods with slight modifications (*Quantifying Food Intake in Caenorhabditis Elegans by Measuring Bacterial Clearance*, 2024), spectrophotometric analysis to measure bacterial consumption was used as a direct measure of food intake in adult worms (day3). Worms were exposed continuously to paroxetine starting from the L1, L3, or YA stage. Before the day of the experiment, ~50 worms were transferred to liquid media containing dead *E. coli* with the appropriate treatments. The dead *E. coli* was prepared, as previously described (Zhu et al., 2020), by heat-killing at 65 °C for 30 min, followed by cooling to room temperature before use. The worms were then incubated at room temperature with gentle shaking for ~30 h to ensure even exposure. After incubation, the remaining bacterial suspension was collected, and the Optical Density (OD) at 600 nm was measured using a spectrophotometer. Food intake was quantified by comparing the OD_{600} of the bacterial suspension across different conditions. The baseline control consisted of dead *E. coli* alone (without worms),

while an additional DMSO control included worms exposed to dead *E. coli* with DMSO but without paroxetine treatment. Bacterial consumption was calculated by determining the difference in OD_{600} after 24 h relative to the DMSO control with worms, providing an accurate estimate of the food intake. Each treatment condition was tested in three independent experiments.

2.4. Fat staining

Fat staining was conducted with minor modifications following a common protocol (Stuhr et al., 2022). using 0.5 % Oil-Red O (ORO). ORO stock solution was prepared by dissolving 500 mg of ORO powder in 100 mL of 100 % isopropanol. Before the experiment, ORO working solution was prepared by diluting the stock solution with water (3:2 ratio) to make 60 % isopropanol. The working solution was filtered with a 0.2 μm syringe filter. Worms were treated from the L1 larval stage with various concentrations of the drugs and then stained at the L4 stage to measure body fat. Images of stained worms were analyzed using ImageJ software for color density. Experiments were performed with approximately 10 animals per replicate and repeated twice.

2.5. Lifespan

Following a previous protocol (*Measuring Caenorhabditis Elegans Life Span on Solid Media*, 2009), Age-synchronized worms were exposed continuously to paroxetine starting from the L1, L3, or YA stage, as standard in *C. elegans* studies. Developmental stages were determined according to the standard life cycle at 20 °C (Dhakal et al., 2021), with larvae reaching L1 at 10–12 h, L3 at 28–30 h, and YA at 40–48 h post-hatching, confirmed morphologically. All treatment groups, including controls, received DMSO from the L1 stage to ensure uniform solvent exposure, even if the specific treatment was introduced later.

Worms were initially cultured in large groups on NGM plates without fluorodeoxyuridine (FUDR). When worms reached the L3 stage, a subset from each group was randomly selected and transferred to the corresponding treatment plates, still without FUDR. Upon reaching the L4/YA

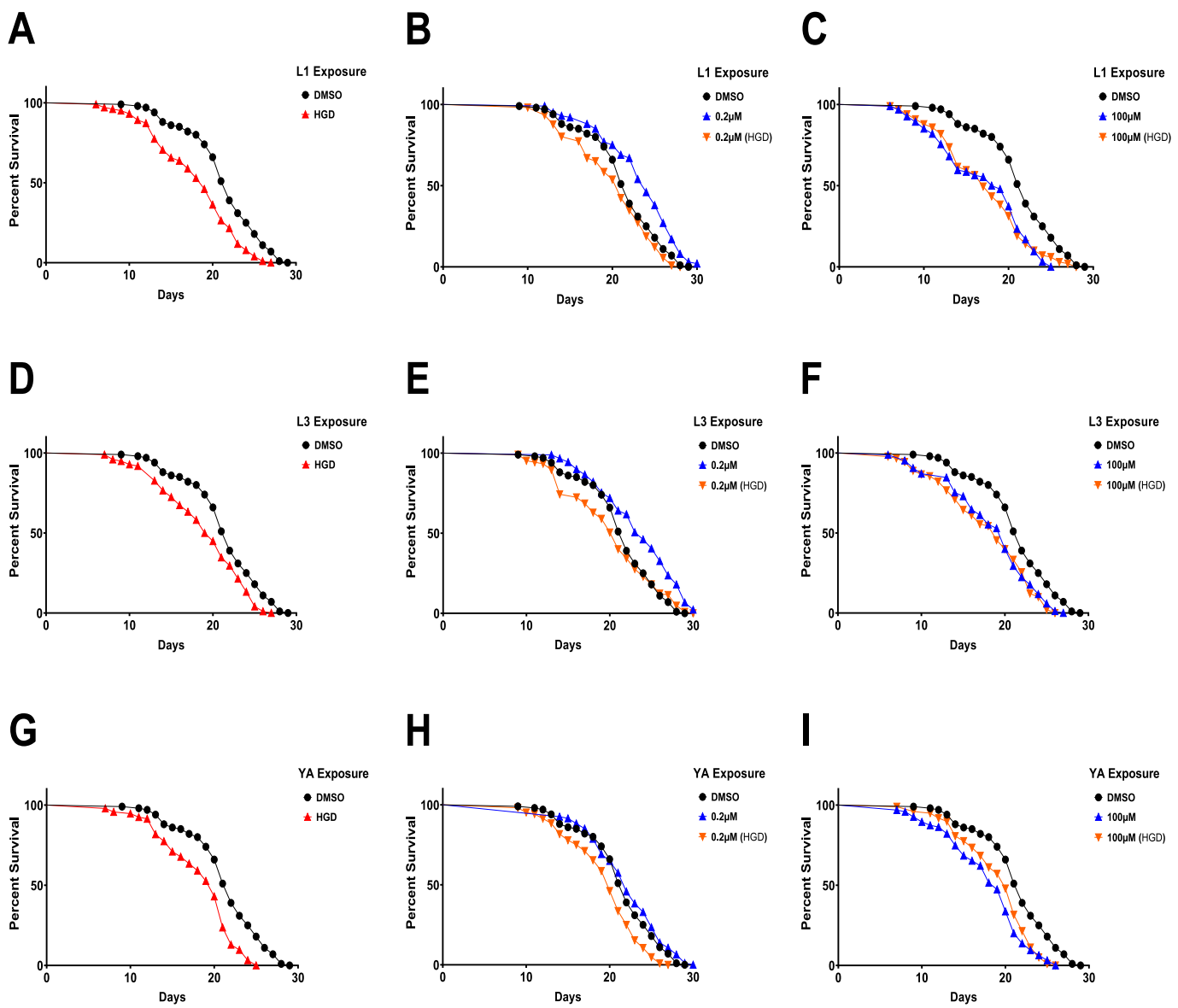

Fig. 1. Survival curves were generated using Kaplan-Meier analysis to assess the impact of paroxetine and HGD on *C. elegans* lifespan. Statistical significance was determined using the log-rank test. DMSO was used as the control condition. Panels (A, D, G) show the effects of HGD alone in worms exposed from the L1, L3, and YA stages, respectively. (B, E, H) and (C, F, I) illustrate the effects of low-dose (0.2 μM) and high-dose (100 μM) paroxetine, respectively, in the presence and absence of HGD at each developmental stage.

stage), approximately 30 worms per replicate were randomly chosen and transferred to FUDR-containing plates (33 μL of 150 mM FUDR per 100 mL NGM) along with the appropriate treatment to prevent reproduction. From this point, worms were monitored daily until death and scored as dead when they failed to respond to gentle stimulation with a platinum pick. The experiment was performed in triplicate.

2.6. Statistical analysis

All data were analyzed using GraphPad Prism 8.0 and JASP statistical software, and results are presented as mean ± SEM. Group differences were assessed using one-way and two-way ANOVA, followed by Tukey's post hoc test to identify specific pairwise differences. For survival analysis, Kaplan-Meier (KM) survival curves were generated and compared using appropriate statistical tests. *P*-values of ≤0.05 were considered statistically significant.

3. Results

3.1. Lifespan

To investigate the prolonging effect of paroxetine on the lifespan of *C. elegans*, we selected two concentrations based on prior studies. The low dose (0.2 μM) has been previously associated with lifespan extension, and the high dose (100 μM) has been reported to reduce survival (Petrascheck et al., 2007; Zhou et al., 2024). These concentrations were chosen to evaluate the contrasting physiological effects of paroxetine under varying dietary and developmental conditions. The data presented in **Panels A, B,** and **C** of Fig. 1 illustrate the effects of paroxetine and a high-glucose diet on lifespan when exposure began at the L1 larval stage. **Panel A** shows that worms exposed to HGD alone experienced a significant reduction in lifespan compared to the DMSO control, indicating that early-life exposure to high-glucose conditions has a detrimental impact on survival. The findings in **Panel B** reveal that low-dose paroxetine extended lifespan in the absence of HGD, but this beneficial effect was completely eliminated when the drug was combined with HGD. Additionally, **Panel C** demonstrates that high-dose paroxetine significantly decreased lifespan, and this negative impact was maintained when HGD was introduced, suggesting a strong interaction between early exposure to both the drug and diet in determining longevity. (See Table 1.)

Panels D, E, and **F** present the results for worms exposed to the treatments starting from the L3 larval stage. As shown in **Panel D**, exposure to HGD alone from the L3 larval stage resulted in a reduction in lifespan, although the impact was marginally less pronounced compared to worms exposed from the L1 stage. Interestingly, **Panel E** demonstrates that low-dose paroxetine treatment slightly extended lifespan in the absence of HGD, but this beneficial effect was eliminated when the drug was combined with the high-glucose diet. Furthermore, **Panel F** illustrates that high-dose paroxetine exposure continued to shorten lifespan, and the addition of HGD further exacerbated this detrimental impact, although the effects were somewhat milder than those observed in the L1-exposed cohort.

Panels **G, H,** and **I** present data from worms exposed to the treatments from the YA stage**. Panel G** shows that HGD alone still shortened lifespan, but the effect was milder than in the earlier exposure groups. **Panel H** demonstrates that low-dose paroxetine had no significant lifespan-extending effect, regardless of HGD presence, suggesting that the beneficial effects observed in earlier stages may be developmentally dependent. **Panel I** shows that high-dose paroxetine continued to decrease lifespan, with the combination of HGD leading to a further reduction in survival, though the effects were again less pronounced than in L1-exposed worms.

To further investigate the physiological effects of paroxetine, we assessed its impact on food intake in *C. elegans* using the same two concentrations (0.2 μM and 100 μM) employed in the lifespan experiments. Pharyngeal pumping rates were measured under different dietary and developmental conditions, as shown in Fig. 2. Each panel represents a different combination of paroxetine dose and developmental timing. **Panel A** shows the effects of 0.2 μM paroxetine with or without HGD starting from the L1 stage. Off-food pumping rates were slightly reduced in all treated groups compared to DMSO. Food intake appeared mildly elevated in worms treated with paroxetine alone or when combined with HGD, although the differences were not statistically significant. On-food rates remained comparable across all groups. **Panel B** displays results from worms exposed to 100 μM paroxetine from the L1 stage. A similar reduction in off-food pumping was noted, but food intake and on-food rates did not markedly differ from the control.

In **Panel C**, worms were treated with 0.2 μM paroxetine from the L3 stage. A modest reduction in off-food pumping was observed, while food intake showed a slight, non-significant increase, particularly in the group exposed to both paroxetine and HGD. On-food pumping rates were stable across all conditions. **Panel D** represents the effects of 100 μM paroxetine beginning at the L3 stage. Again, a decrease in off-food pumping was seen, especially when paroxetine was combined with HGD. On the other hand, food intake and on-food rates remained largely unchanged. **Panels E** and **F** reflect data from young adult exposure. In **Panel E**, worms treated with 0.2 μM paroxetine exhibited mildly reduced off-food pumping rates. Food intake showed a trend toward elevation in the treated groups, especially with HGD, though not statistically significant. On-food rates remained similar among all groups. Finally, **Panel F** presents results for 100 μM paroxetine initiated at the young adult stage. Off-food rates were reduced across treatment groups, but food intake and on-food pumping remained relatively consistent with controls.

Since we did not observe strong or consistent changes in pharyngeal pumping rates, we decided to measure bacterial consumption using OD_{600} as a more integrative measure of food intake. As shown in Fig. 3, **Panel A** presents the data for worms exposed to 0.2 μM paroxetine from the L1 stage. A clear reduction in bacterial density was observed following treatment with paroxetine in the presence and absence of HGD, suggesting an increase in food intake under these conditions. **Panel B** shows the corresponding results for 100 μM paroxetine from the L1 stage. Unlike the lower dose, this treatment did not significantly alter OD_{600} values compared to the control (DMSO). Both the paroxetine-only and the combined HGD treatment groups exhibited similar bacterial densities to their respective controls. **Panel C** highlights the effect of 0.2 μM paroxetine initiated at the L3 stage. A modest, non-significant reduction in OD_{600} was noted with paroxetine alone. However, worms

Table 1
Lifespan of *C. elegans* under different experimental conditions. Mean lifespan (± SEM) from three independent replicates ($N \geq 30$ worms per group). Statistical analysis was performed using one-way ANOVA with Tukey's post hoc test ($p < 0.05$); (*) indicates significant difference from control.

Treatments		Median	Mean	SEM	95 Confidence Interval Mean Lower	Upper
DMSO		22	21.31	0.436	20.44	22.18
	HGD	19	18.01*	0.489	17.04	18.98
	0.2 μM	24	23.28*	0.453	22.38	24.18
L1 Exposure	0.2 μM(HGD)	21	20.02	0.451	19.13	20.91
	100 μM	18.5	17.06*	0.554	15.96	18.16
	100 μM(HGD)	17	17.12*	0.524	16.08	18.16
	HGD	19.5	18.86*	0.501	17.86	22.18
	0.2 μM	24	23.43*	0.502	22.43	19.85
L3 Exposure	0.2 μM(HGD)	21	20.06	0.522	19.02	24.42
	100 μM	20	18.4*	0.562	17.28	21.09
	100 μM(HGD)	19	17.84*	0.571	16.71	19.52
	HGD	20	18.33*	0.457	17.42	19.24
	0.2 μM	22	22	0.436	21.13	22.87
YA Exposure	0.2 μM(HGD)	20	19.34*	0.426	18.49	20.18
	100 μM	19	17.86*	0.488	16.89	18.83
	100 μM(HGD)	20	19.03*	0.426	18.18	19.88

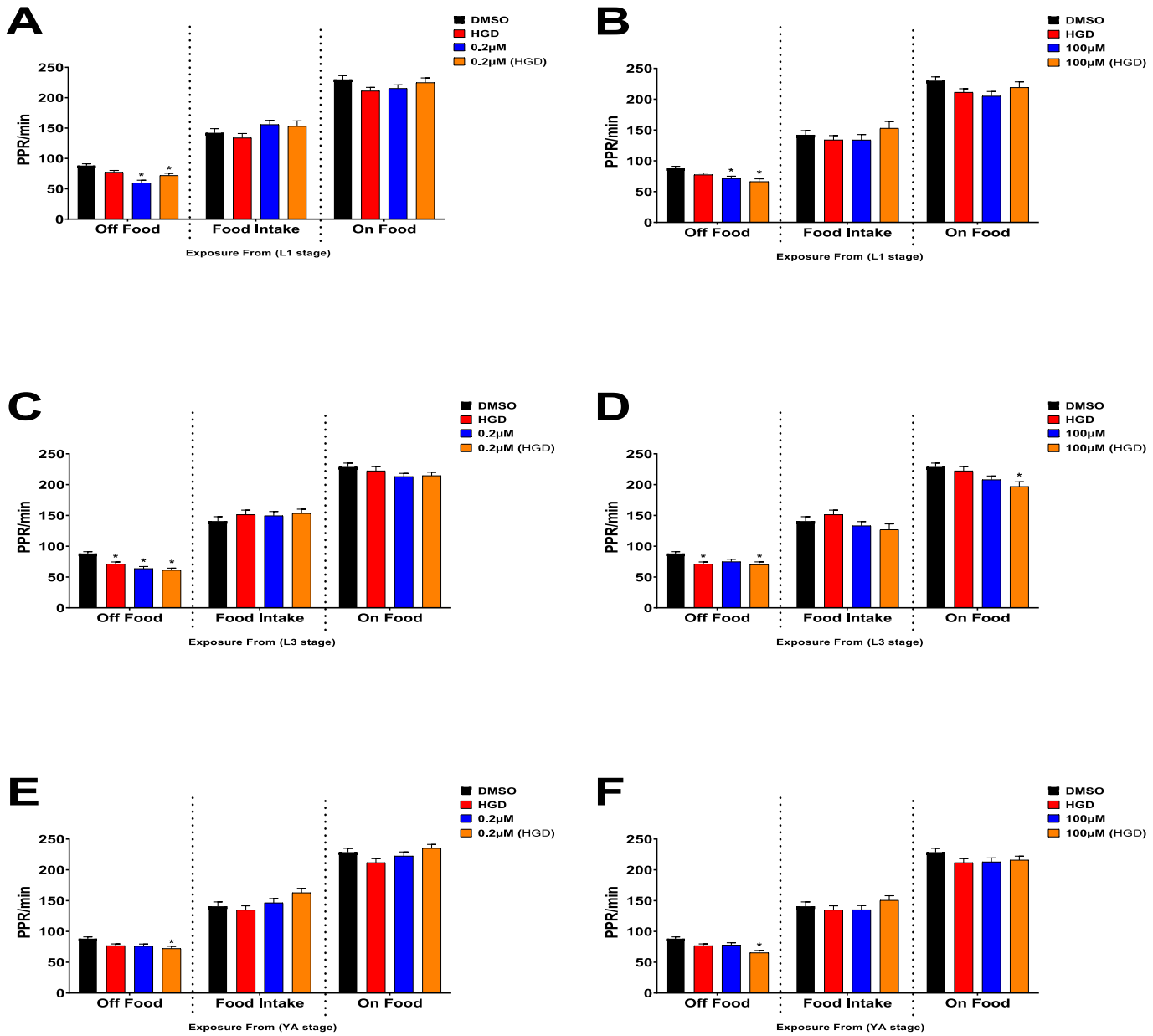

Fig. 2. Pharyngeal pumping rate (PPR; pumps min^{-1}) in *C. elegans* following paroxetine exposure. Worms were exposed to 0.2 μM or 100 μM paroxetine from the L1, L3, or YA stages under a normal or high-glucose diet (HGD). PPR was measured Off Food, during Food Intake, and On Food. Panels A–B: exposure from L1; C–D: from L3; E–F: from young adults. Bars represent mean ± SEM from three independent biological replicates (10 worms per group in each replicate). Statistical analysis was performed using one-way ANOVA with Tukey's post hoc test; * $p < 0.05$.

exposed to the combined treatment with HGD displayed a significant reduction in OD_{600}. **Panel D** presents the data for 100 μM paroxetine exposure from the L3 stage. Similar to the results from the L1 exposure, no significant changes in bacterial density were observed across any of the treatment groups. This suggests a limited effect of the higher paroxetine dose on bacterial intake at this developmental stage.

Panels E and **F** represent the responses of worms treated from the YA stage with 0.2 μM and 100 μM paroxetine, respectively. In both panels, the OD_{600} values remained consistent across all groups, indicating no significant alterations in bacterial consumption. These findings suggest that late exposure to paroxetine does not affect feeding behavior, at least as assessed by this method. Overall, these results demonstrate that the effects of paroxetine on bacterial consumption in *C. elegans* are dose-dependent and influenced by the timing of exposure, with the most pronounced effects occurring when the drug is introduced during early developmental stages and combined with a high-glucose diet.

3.2. Role of specific genes in paroxetine-induced food intake

Since the most pronounced increase in food intake was observed at 0.2 μM paroxetine when exposure began at the L1 stage, we selected this condition to further investigate the underlying genetic pathways involved in mediating the drug's effect on feeding behavior. Therefore, we used various mutant strains along with N2 wild-type to investigate further the observed effects of paroxetine on bacterial consumption. OD_{600} measurements were used to assess bacterial consumption, providing a direct measure of food intake. Each panel of Fig. 4 represents a different genetic background, highlighting potential pathway involvement in the drug's effect on feeding behavior. **Panel A** presents the results for the wild-type (N2) strain, showing that worms treated with paroxetine exhibited reduced bacterial consumption compared to the DMSO control. This suggests that paroxetine enhances feeding behavior in *C. elegans*. **Panels B** and **C** display results for *daf-2* and *daf-*

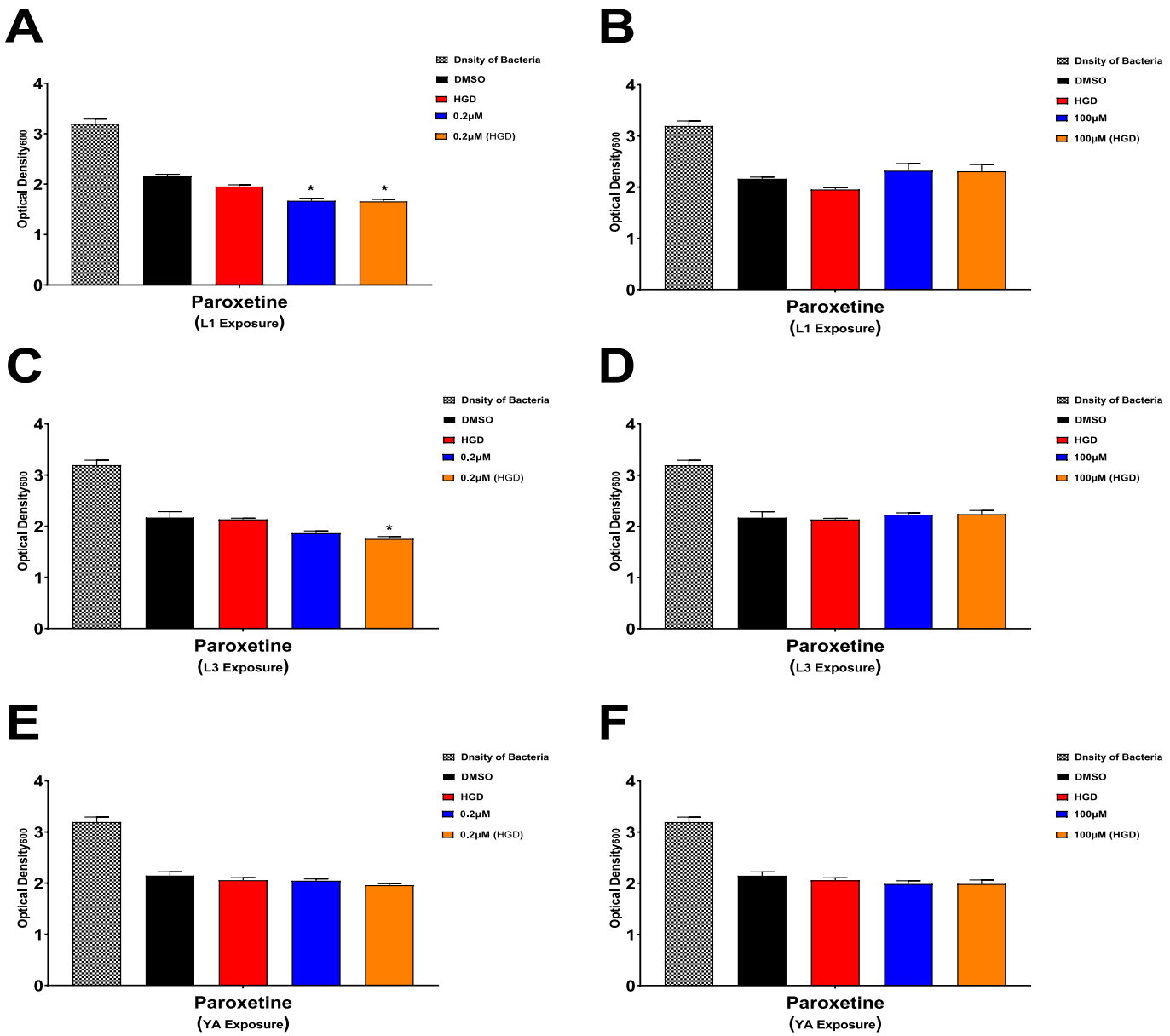

Fig. 3. Bacterial consumption in *C. elegans* following paroxetine exposure at different developmental stages. Panels A–B: exposure from the L1 stage; C–D: from the L3 stage; E–F: from the young adult (YA) stage. Worms were treated with 0.2 μM or 100 μM paroxetine, and bacterial consumption was quantified as the optical density (OD_{600}) of remaining *E. coli* OP50. Bars represent mean ± SEM from three independent biological replicates (~50 worms per group in each replicate). Statistical analysis: one-way ANOVA with Tukey's post hoc test; $p < 0.05$.

16 mutants, respectively, both of which showed a similar trend of decreased bacterial consumption following drug exposure. The consistency between these mutants and the wild type suggests that the insulin-like signaling pathway does not significantly influence the observed feeding behavior. **Panels D** through **I** illustrate bacterial consumption in *ser-1, ser-2, ser-3, ser-4, ser-5, and ser-7* mutants, which are defective in different serotonergic receptors. Across all these mutants, paroxetine exposure resulted in a reduction in bacterial consumption comparable to the wild-type worms. This indicates that the serotonin receptors tested do not appear to mediate the drug's effect on feeding behavior. The similarity in response across these mutants suggests that serotonergic signaling may not be a primary pathway through which paroxetine alters food intake in *C. elegans*. **Panel J** shows the results for *tbh-1* mutants, which have defects in a different aspect of octopaminergic signaling pathway. Similar to the wild-type, these mutants displayed reduced bacterial consumption following paroxetine treatment. **Panels K** and **L** depict the results for *dop-1* and *dop-2* mutants, which are impaired in different aspects of the dopaminergic signaling pathway. These mutants exhibited decreased bacterial consumption upon drug exposure, similar to the wild type and other strains tested. This suggests that dopaminergic pathways do not play a major role in mediating paroxetine-induced feeding suppression.

Since the effect of paroxetine on feeding was most pronounced at low doses and specifically when exposure began at early developmental stages, we decided to evaluate if it would affect the body fat under the same conditions. **Panel A** in Fig. 5 illustrates the results for worms exposed to 0.2 μM paroxetine from the L1 stage. In response to paroxetine alone and in combination with HGD, fat content remained largely unchanged. The levels were comparable to those in both DMSO and HGD-only controls. **Panel B** shows the fat content in worms treated with a higher dose of paroxetine (100 μM) under the same exposure conditions. Similar to the lower dose, fat accumulation did not differ notably

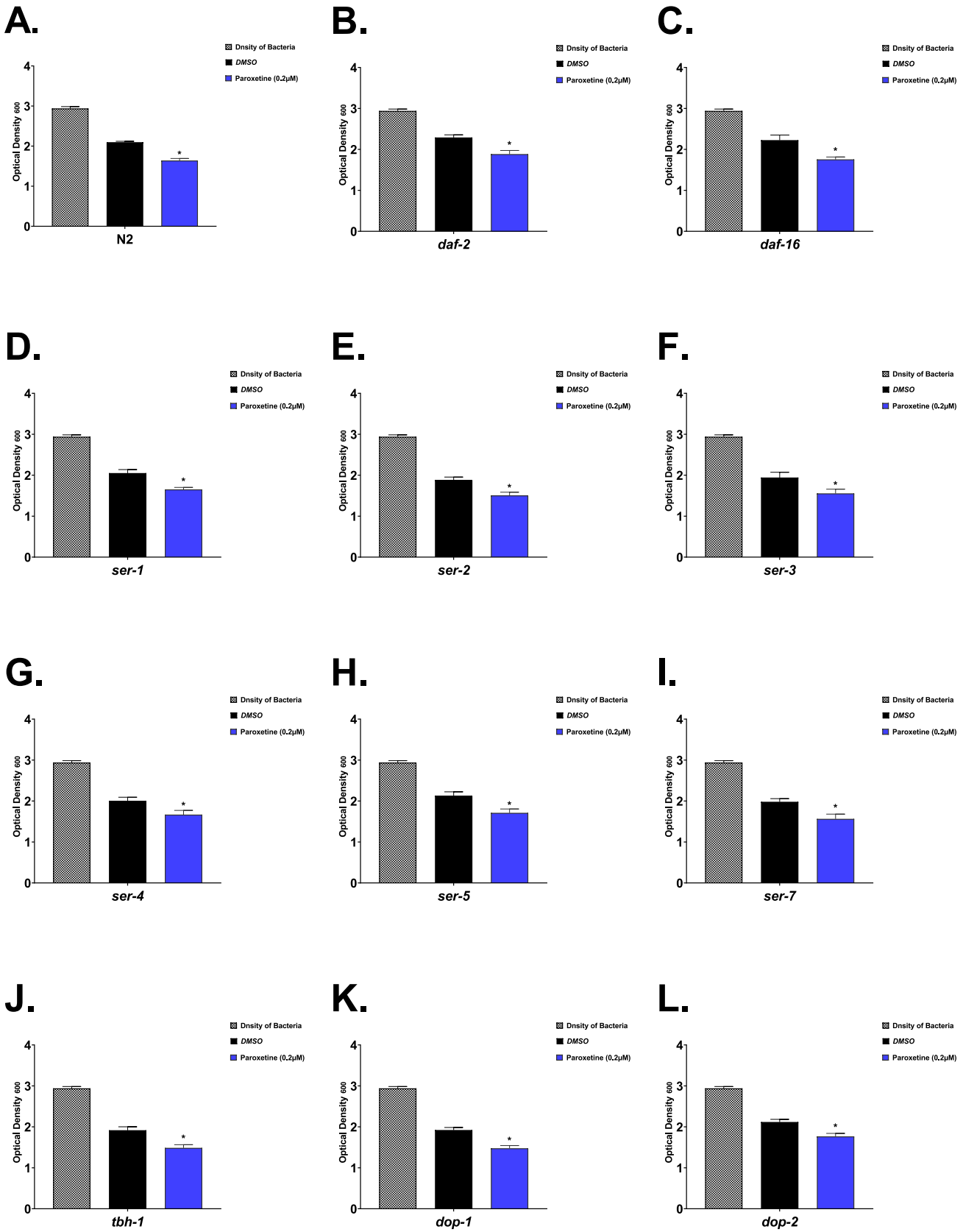

Fig. 4. Bacterial consumption OD_{600} in wild-type and mutant *C. elegans* strains following exposure to 0.2 μM paroxetine from the L1 stage. Panels A–L show OD600-based bacterial density measurements for wild-type N2 (A), *daf-2* (B), *daf-16* (C), *ser-1* (D), *ser-2* (E), *ser-3* (F), *ser-4* (G), *ser-5* (H), *ser-7* (I), *tbh-1* (J), *dop-1* (K), and *dop-2* (L). Data are presented as mean ± SEM. Statistical analysis was performed using one-way ANOVA followed by Tukey's post hoc test; $p < 0.05$ was considered significant (* $p < 0.05$).

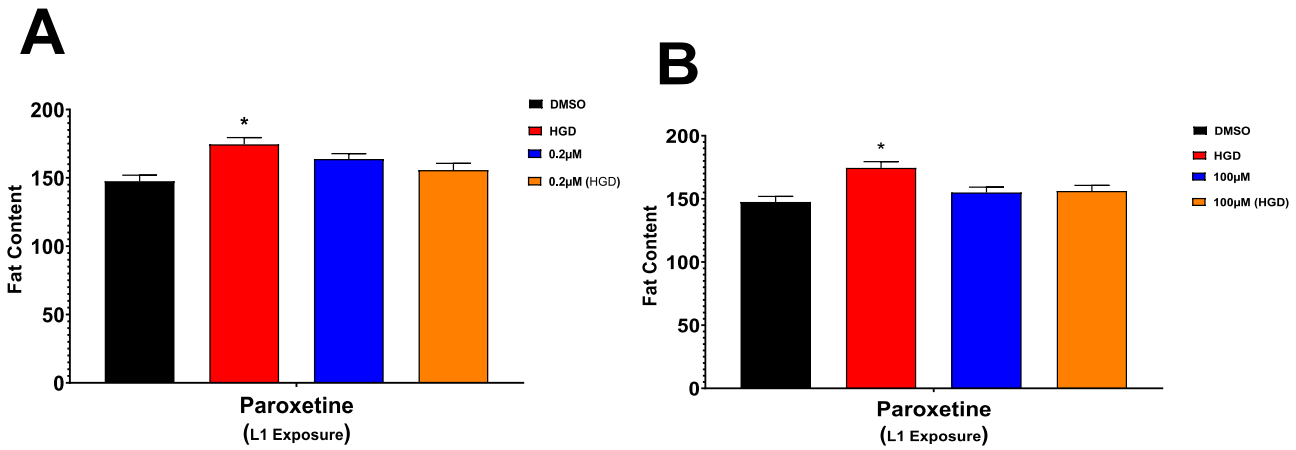

Fig. 5. Effects of paroxetine on fat content in *C. elegans* following exposure from the L1 stage. **Panel A** shows the fat content in worms treated with 0.2 μM paroxetine, with or without a high-glucose diet (HGD), compared to DMSO and HGD-only controls. **Panel B** presents similar conditions for worms treated with 100 μM paroxetine. Data represent mean ± SEM from 2 independent biological replicates (10 worms per group per replicate). Statistical analysis was performed using one-way ANOVA followed by Tukey's post hoc test; $p < 0.05$ was considered significant (* $p < 0.05$).

between groups. This suggests that even though paroxetine may increase bacterial consumption, it does not result in measurable changes in fat storage under these conditions.

4. Discussion

In this study, we used the *C. elegans* model to investigate paroxetine effects on several physiological parameters, such as lifespan, feeding behavior, and fat metabolism under various dietary conditions and exposure timings. Our results reveal intricate interactions between drug dose, time of exposure, and dietary composition. These interactions significantly influence physiological outcomes in this model organism. These effects are likely mediated through serotonergic and metabolic pathways, aligning with paroxetine's known actions in both nematodes and mammals. Such findings provide new insights into how SSRIs may function beyond neurotransmitter modulation and highlight important clinical implications.

Our investigation showed that paroxetine exhibits clear dose-dependent effects on the lifespan of *C. elegans*. Low doses of the drug (0.2 μM) extended longevity, whereas high doses (100 μM) reduced it. These hormetic responses of the drug on the lifespan of worms are in line with previous research (Petrascheck et al., 2007; Zhou et al., 2024). These responses to paroxetine may reflect an adaptive stress response mechanism. At low doses, paroxetine might induce mild cellular stress that activates protective pathways. This resembles the concept of mitohormesis, where mild mitochondrial stress triggers compensatory responses that enhance overall organismal fitness. Several compounds in *C. elegans* show this phenomenon, including metformin and certain antioxidants (De Haes et al., 2014; Yu et al., 2022). Conversely, high doses of paroxetine likely overwhelm cellular homeostatic mechanisms, leading to toxicity and a reduced lifespan. The molecular pathways mediating these effects remain to be fully elucidated, and further research is needed.

Moreover, the concentration of the drug was not the only factor that played a role in the observed effect of the drug. The time of exposure played a significant role in modulating the effect of the drug on lifespan and other physiological parameters. The lifespan-extending effects of low-dose paroxetine were most substantial when exposure began at early developmental stages (L1) and then gradually diminished when treatment started at later stages (L3 and young adult). Previous studies in *C. elegans* have highlighted that the timing of exposure is a crucial determinant of how certain compounds influence lifespan and other physiological outcomes (Munasinghe et al., 2021). Our study extends these observations by demonstrating that exposure timing significantly modulates these effects. Early developmental exposure appears critical for lifespan extension to occur.

Interestingly, the effect of paroxetine on lifespan was also influenced by the dietary context, which played a significant role in modulating the observed outcomes. In this study, we used 73 mM glucose as a mid-range concentration within the commonly employed 50–100 mM window for *C. elegans* HGD models, selected to reliably induce diet-related metabolic stress while avoiding osmotic or developmental effects (Lee et al., 2009; Schlotterer et al., 2009; Alcántar-Fernández et al., 2018). First, our findings revealed that the addition of glucose alone (HGD) to the typical bacterial diet reduces the lifespan of worms compared to control conditions, confirming previous research findings (Lee et al., 2009; Schlotterer et al., 2009; Choi, 2011; Almotayri et al., 2021). This lifespan reduction in response to HGD is attributed to increased reactive oxygen species (ROS) generation by glucose metabolism, leading to oxidative damage of cellular components, including proteins, lipids, and nucleic acids (Alcántar-Fernández et al., 2018). Second, when we examined the combined effect of paroxetine treatment and HGD across different developmental stages (L1, L3, and young adult), we observed different responses to the drug. Exposure to a high-glucose diet eliminated the beneficial effects of low-dose paroxetine on lifespan while making the detrimental effects of high-dose treatment worse. Prior research has shown that the dietary context is a critical factor in determining the effects of specific chemicals on *C. elegans (*Almotayri et al., *2021)*. These findings emphasize the importance of time of exposure and nutritional context when evaluating SSRI effects on physiological outcomes.

The mechanistic interplay between glucose metabolism and paroxetine action may explain our observations. While glucose generates ROS in both human and *C. elegans* cells (Alcántar-Fernández et al., 2018; Ha and Lee, 2000), low-dose paroxetine may promote longevity through mechanisms that may involve stress resistance pathways. The opposing biochemical effects of these interventions could explain why HGD counteracts paroxetine's beneficial effects. This suggests that the translational relevance of *C. elegans* studies on paroxetine may be limited by the substantial differences in typical dietary composition between nematodes and humans. The standard laboratory diet for *C. elegans* consists primarily of *E. coli* bacteria, which provides a macronutrient profile of approximately 55 % protein, 23 % nucleic acids, 9 % lipids, and only 3 % carbohydrates in the form of glycogen (Smith, 1992). This contrasts sharply with contemporary human dietary patterns, where carbohydrates often constitute 45–65 % of daily caloric intake, according to many national dietary guidelines (Mayo Clinic Staff, 2025; Slavin and Carlson, 2014; Buyken et al., 2018). Nutritional surveys conducted over recent decades indicate a substantial increase in

processed carbohydrate and added sugar consumption among populations in developed countries (Shan et al., 2019; Lee et al., 2018). Furthermore, the relationship between psychotropic medications and dietary patterns presents another layer of complexity. Clinical observations have documented altered eating behaviors in patients taking SSRIs and other psychotropic medications, with many reporting heightened appetite and specific cravings for carbohydrate-rich and sweet foods (Yeragani et al., 1988). This suggests that medications prescribed to improve mental health may inadvertently promote dietary patterns that undermine their therapeutic efficacy.

Dietary composition is increasingly recognized as a key factor influencing neuropsychiatric treatment outcomes. Low-carbohydrate and ketogenic diets have shown cognitive and neurological benefits, including improved memory (Krikorian et al., 2012) and reduced symptoms in epilepsy (Borowicz-Reutt et al., 2024; Martin-McGill et al., 2020), Alzheimer's (Henderson, 2008), Parkinson's (Price and Ruppar, 2023), and other neurological disorders (Chinna-Meyyappan et al., 2023; Har-Even et al., 2021). Clinical evidence also suggests that dietary interventions can enhance antidepressant efficacy (Grochowska and Przeliorz, 2022; Mocking et al., 2015; Decker et al., 2025). Consistent with these findings, our lifespan data highlight the importance of personalized SSRI therapy that considers diet, developmental history, and metabolism. Future studies should examine whether reducing refined carbohydrate intake can improve paroxetine efficacy and clarify dose- and stage-dependent effects under varying nutritional conditions.

Our investigation into the effects of paroxetine on feeding behavior revealed an intriguing paradox. The evaluation of feeding was conducted by employing two different techniques. The first technique was calculating the pharyngeal pumping rates, and the other one was measuring the bacterial consumption rates by measuring OD_{600}. The pharyngeal pumping rates showed only modest changes across treatment conditions, whereas bacterial consumption measurements indicated that low-dose paroxetine (0.2 μM) significantly increased food intake, particularly when combined with a high-glucose diet. This effect was most pronounced when exposure began at early developmental stages (L1 and L3) and was absent when treatment started at the young adult stage. The variations in food intake between the two techniques imply that paroxetine may affect metabolic efficiency or nutritional absorption instead of only the motor activity of feeding. Although pharyngeal pumping rate represents short-term behavioral changes, bacterial consumption provides a more integrated picture of actual food intake over time, which would help to explain the more significant impacts found with the latter approach. This discrepancy highlights the importance of using multiple complementary approaches to assess behavioral and physiological changes in response to pharmacological treatments.

Given that paroxetine's impact on feeding was most pronounced at low doses (0.2 μM) and with exposure commencing at early developmental stages, we determined to assess its potential impact on body fat under the same circumstances. Surprisingly, despite this increased food consumption, we observed no corresponding increase in fat accumulation. This result implies that the drug may affect food intake without affecting the mechanisms involved in fat synthesis. These results are consistent with prior research, confirming the fact that *C. elegans* is governed by distinct mechanisms for the regulation of lipid and food intake (Srinivasan et al., 2008). This metabolic contradiction contrasts with the common clinical observation that SSRI treatment in humans is often associated with weight gain and increased adiposity, particularly with long-term use (Bansal et al., 2022; Cartwright et al., 2016). Paroxetine, specifically, has been linked to greater weight gain compared to other SSRIs in clinical studies (Nevels et al., 2016). The discrepancy between our findings in *C. elegans* and human clinical observations highlights the complexity of SSRI effects on metabolism and the potential limitations of direct cross-species extrapolation. Species differences in serotonergic system organization, metabolic regulation, and drug pharmacokinetics may contribute to these divergent outcomes. Alternatively, the duration of exposure in our study may not have been sufficient to capture long-term metabolic adaptations that could eventually lead to fat accumulation.

To further investigate the mechanisms by which paroxetine affects feeding behavior, we assessed bacterial consumption in a series of *C. elegans* mutants with defects in serotonergic, dopaminergic, insulin-like, and octopaminergic signaling pathways. This selection was guided by a recent study showing that paroxetine extends lifespan via the *ser-7* and *dop-4*–IIS axis (Zhou et al., 2024), suggesting a role for these pathways in mediating its physiological effects. Surprisingly, paroxetine reduced food intake across all tested mutants, including serotonin receptor mutants (*ser-1* to *ser-7*), insulin pathway mutants (*daf-2* and *daf-16*), dopamine receptor mutants (*dop1* to *dop-4*), and the octopaminergic mutant (*tbh-1*). These persistent effects indicate that the impact of paroxetine on feeding behavior is not dependent exclusively on the pathway we examined and could involve other or intersecting pathways. Altogether, these findings point toward broader or redundant mechanisms of action, potentially involving fundamental cellular processes beyond classical monoaminergic signaling.

Despite these compelling findings, several limitations need to be acknowledged. The first limitation is using commercially available paroxetine tablets, which include pharmaceutical excipients that could potentially affect physiological outcomes. However, this approach was chosen to reflect real-world clinical exposure and the conditions under which paroxetine is typically administered in humans. Moreover, the concentration of the active ingredient was carefully adjusted to match previously reported effective doses in *C. elegans*, minimizing the likelihood that excipients affected the results. Nonetheless, to ensure full reproducibility and eliminate potential confounding effects, future studies should confirm these findings using purified, laboratory-grade paroxetine. Another limitation is that the concentrations of paroxetine used in our experiments do not directly reflect therapeutically relevant levels in humans due to species-specific differences in drug absorption and metabolism. Therefore, the results should be interpreted with caution when considering their relevance to human dosing. Moreover, while we focused on key developmental stages and selected physiological markers, a broader dose-response analysis and extended follow-up could help reveal potential long-term or even transgenerational effects. Finally, while *C. elegans* remains a powerful system for uncovering conserved biological processes, caution is warranted when translating these results to humans, especially considering the structural and functional differences in serotonin pathways and metabolic regulation. Still, our findings highlight promising directions for future research, including the need to explore SSRI-diet interactions in mammalian systems and consider how developmental and nutritional factors influence antidepressant outcomes.

5. Conclusion

The study offers new insights into how paroxetine influences lifespan, feeding behavior, and metabolism in *C. elegans*, with results pointing to a complex interplay between drug dose, timing of exposure, and dietary context. The drug, under certain conditions, offers many benefits that were completely abolished when the worms were fed a high-glucose diet. This strong interaction between drug and diet suggests that nutritional context plays a critical role in shaping the physiological impact of SSRIs. Additionally, we found that paroxetine increased food intake without a corresponding rise in fat accumulation, indicating a possible state of metabolic uncoupling. This raises important questions about how SSRIs influence energy balance, especially given that these effects persisted across various mutant strains, suggesting the involvement of non-canonical or redundant pathways beyond traditional serotonin signaling.

CRediT authorship contribution statement

Abdullah M. Almotayri: Writing – review & editing, Writing – original draft, Supervision, Methodology, Data curation, Conceptualization. **Ali H. Alghamdi:** Writing – review & editing, Visualization, Conceptualization. **K. ELSHERBINY:** Writing – review & editing, Validation, Formal analysis, Data curation, Conceptualization. **Ali A. Hroobi:** Writing – review & editing, Visualization, Validation. **Nourah M. Almimoni:** Writing – review & editing, Visualization, Validation. **Hussam A. Althagafi:** Writing – review & editing, Visualization, Validation, Conceptualization. **Tariq Saeed Alghamdi:** Writing – review & editing, Visualization, Validation. **Fatehia N. Gharsan:** Writing – review & editing, Visualization. **Mariam S. Alghamdi:** Writing – review & editing, Visualization. **Abdullah A.A. Alghamdi:** Writing – review & editing, Visualization, Validation, Conceptualization.

Funding

The authors received no financial support from any organization, agency, or institution for the research, authorship, or publication of this work.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have influenced the work reported in this paper.

Data availability

The datasets generated and analyzed during this study are available from the corresponding author upon reasonable request.

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